CLASS 8 — SCIENCE · NCERT-TYPE UNIT TEST SET 3

Chapters 10–13: **Light: Mirrors and Lenses, Keeping Time with the Skies, How Nature Works in Harmony, Our Home: Earth, a Unique Life-sustaining Planet** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. In a food chain, the first trophic level is occupied by: 1

(a) decomposers **(b)** herbivores **(c)** producers **(d)** carnivores

2. The lowest layer of the atmosphere in which weather occurs is the: 1

(a) mesosphere **(b)** troposphere **(c)** thermosphere **(d)** stratosphere

3. A dentist uses a: 1

(a) plane mirror **(b)** concave mirror **(c)** convex mirror **(d)** no mirror

4. A year of the Gregorian calendar has: 1

(a) 400 days **(b)** 354 days

(c) 365 days (366 in a leap year) **(d)** 360 days

5. Which gas do plants release during photosynthesis? 1

(a) methane **(b)** carbon dioxide **(c)** oxygen **(d)** nitrogen

6. The water that seeps into the ground and fills spaces in rocks is: 1

(a) surface water **(b)** vapour **(c)** groundwater **(d)** rain

7. The angle of incidence is measured between the incident ray and the: 1

(a) reflected ray **(b)** mirror surface **(c)** horizontal **(d)** normal

8. A leap year has: 1

(a) 360 days **(b)** 366 days **(c)** 365 days **(d)** 364 days

9. Organisms that make their own food are called: 1

(a) consumers **(b)** parasites **(c)** producers **(d)** decomposers

10. The innermost layer of the Earth is the: 1

(a) mantle **(b)** crust **(c)** outer core **(d)** inner core

11. The number of images formed by two plane mirrors placed at 90 degrees to each other is: 1

(a) 1 **(b)** 3 **(c)** 2 **(d)** 4

12. When the Moon is not visible, it is the: 1

(a) full moon **(b)** gibbous moon **(c)** new moon **(d)** half moon

13. A food web is: 1

(a) a plant **(b)** many interlinked food chains
(c) a type of spider web **(d)** a single chain

14. The ozone layer is present in the: 1

(a) mesosphere **(b)** exosphere **(c)** troposphere **(d)** stratosphere

15. A shaving mirror is a: 1

(a) plane mirror **(b)** concave mirror **(c)** convex mirror **(d)** spherical lens

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. What is biodiversity? Why is it important? 2

17. State two uses of the atmosphere for life. 2

18. Differentiate between regular and diffuse reflection. 2

19. What causes day and night? 2

20. Differentiate between biotic and abiotic components with two examples each. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Explain the role of producers, consumers and decomposers in an ecosystem. 3

22. Describe the composition of air with approximate percentages. 3

23. Name the parts of a spherical mirror: pole, centre of curvature, principal axis, focus. Define any two. 3

24. What are the effects of the Earth's rotation? List three. 3

OR

State three effects of deforestation. 3

25. Explain why only a small part of water on Earth is available as fresh water. 3

OR

Explain how a convex lens forms a real image of a distant object (such as the Sun) on a screen. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Describe the characteristics of the image formed in a plane mirror. Two plane mirrors are inclined at 90 degrees. Find the number of images and explain. 5

OR

Describe the phases of the Moon through a month with reasons and the names for waxing and waning phases. 5

27. Describe food chains and food webs with two examples and explain why a food web is more stable than a food chain. 5

OR

Describe the layers of the atmosphere with height and a feature of each. 5

END OF QUESTION PAPER — Total: 50 Marks